
A Robot Policy Designed as a Weight-Space Artifact

Exact Tensor Structure in a Capable VLA

Rational conversion, reduced-Gram analysis, and exact weight access

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Abstract

1 Tensor decomposition rewrites a multidimensional table as a small set of low-rank
2 factors. In a neural network, those factors can expose which combinations of inputs
3 and outputs are encoded by the weights. Conventional nonlinear operations make
4 such a table difficult to obtain directly. We introduce χ -VLA, a rational vision-
5 language-action model family with a 19.2M convolutional policy and a 20.1M
6 ViT policy, each containing only bilinear, linear, and rational learned operations.
7 Its checkpoint therefore specifies an explicit rational tensor program rather than
8 only an opaque executable function. A runtime operator audit of the 20.1M ViT
9 checkpoint and local reconstructions verify its operator-level form. We derive
10 an exact weight-based decomposition of each softmax-free attention layer and
11 reconstruct all 12 modules in each of three checkpoints with worst relative error
12 2.56×10^{-7} . An $O(d^2)$ reduced Gram avoids materializing the coefficient tensor.
13 In a seed-0 ViT checkpoint, all 36 primary block-head tests favor the rank-128
14 weight-derived subspace over fixed random controls. Three historical aggregate
15 records for a convolutional instantiation of the same rational model family report
16 $93.7\% \pm 0.8\%$ on LIBERO-Object. The three released convolutional checkpoint
17 files named in those historical aggregate records also pass an independent 4,608-
18 case joint-attention certificate with worst relative error 6.36×10^{-16} . We therefore
19 establish a capable model family with exact layerwise structure derived from
20 trained coefficients. We do not yet provide an exact end-to-end decomposition of
21 the complete policy.

22 1 Introduction

23 Neural checkpoints are usually treated as opaque containers for functions. We can run them and
24 collect their activations, but the weights alone rarely provide a readable map from input features to
25 outputs. A post-hoc probe may reveal that an activation correlates with behavior, yet it does not show
26 how the weights construct that feature.

27 A tensor is a multidimensional table, and tensor decomposition is the analogue of low-rank matrix
28 factorization for such tables. When a network uses only additions, multiplications, and ratios
29 of polynomials, its weights can be expanded into coefficient tensors whose entries record input
30 interactions. Decomposition ranks simpler factors inside those coefficients. The intended benefit
31 is an inspectable model artifact: candidate structure comes from the trained parameters. By fully
32 tensor-decomposable, we mean that every learned block preserves this algebraic form. This is an
33 operator-closure claim, not a claim that we materialize one compact end-to-end coefficient tensor. Its
34 degree and representation size grow rapidly with depth. For example, one coefficient can connect
35 an image coordinate, an instruction coordinate, and an action coordinate. Factoring the coefficient

table ranks high-energy cross-modal interactions directly from the checkpoint without first fitting an activation probe. Those factors are candidate structure, not automatically human-readable concepts or causal explanations. Softmax, ordinary activations, and square-root normalization obstruct this direct conversion in conventional networks. Bilinear language and vision models suggest that changing the primitives can recover it [3, 4]. Whether the resulting artifact can still control a robot is unresolved.

We study two linked questions:

Can the learned weights of a specialist VLA specify an explicit rational tensor program, and can exact weight analysis coexist with strong closed-loop control?

Our contributions are:

1. We construct a complete VLA from tensor-compatible primitives and runtime-audit its 20.1M ViT instantiation. A rational normalization supplies input-specific scaling without leaving the rational function class.
2. We derive exact weight-based structure through each softmax-free attention layer and an $O(d^2)$ reduced-Gram calculation that avoids materializing its high-order coefficient tensor.
3. We test exact attention subspaces against fixed equal-rank random controls. The weight-derived subspaces win all 36 primary block-head comparisons in a seed-0 checkpoint.
4. We report three-seed historical aggregate records for a convolutional instantiation of the same rational family, reaching $93.7\% \pm 0.8\%$ on LIBERO-Object.

Artifact query	Returned object	Current guarantee
Runtime graph census	Operator inventory and module identities	Membership for the audited 20.1M ViT checkpoint
Attention-coefficient query	Reduced Gram and leading directions	Exact layerwise construction from saved weights
Numerical replay	Per-case residuals and frozen gates	Independent implementation check on recorded inputs
Artifact integrity	Result and source digests	Immutable SHA-256 manifest and one-command verification

Table 1: **Checkpoint interface as a data artifact.** The trained checkpoint supports structured queries beyond ordinary forward execution. Guarantees remain layerwise and scoped to the named artifacts.

This paper does not claim current state of the art. It also does not claim that comparable mechanisms are impossible to obtain from conventional policies. The empirical question is whether the algebraic policy makes exact weight structure directly accessible.

2 A rational tensor robot policy

2.1 Tensor-compatible primitives

Each primitive replaces a familiar transformer component while keeping explicit coefficients. Linear maps contribute first-order terms, while multiplying learned projections creates higher-order interactions that remain determined by the weights.

Homogeneous coordinate. We write $\bar{x} = [1; x]$. The constant coordinate allows a multiplicative layer to represent bias, linear, and quadratic terms in one contraction.

Bilinear feed-forward block. For $L, R \in \mathbb{R}^{r \times (d+1)}$ and $D \in \mathbb{R}^{d \times r}$,

$$\text{BFFN}(x) = D[(L\bar{x}) \odot (R\bar{x})], \quad y_o = \sum_{i,j} \left(\sum_{\rho} D_{o\rho} L_{\rho i} R_{\rho j} \right) \bar{x}_i \bar{x}_j. \quad (1)$$

The inner sum is a table indexed by one output and two input coordinates. The three matrices give its CP factorization, the tensor analogue of a low-rank matrix factorization.

67 **Softmax-free bilinear attention.** Each head forms two query-key systems and one value stream:

$$Y = C [M \odot (Q_1 K_1^\top) \odot (Q_2 K_2^\top) / s] V. \quad (2)$$

68 Here M is a fixed causal mask, C is fixed row scaling, and s is fixed. All learned query, key,
69 value, and output maps are linear. Equation 2 is therefore a polynomial in its input activations, with
70 interaction coefficients determined by the weights.

71 **Rational per-instance normalization.** RMS normalization contains an input-dependent square
72 root. Let $m(x) = d^{-1} \sum_{j=1}^d x_j^2 + \epsilon$, and let $s_0 > 0$ be the frozen running mean-square scale. We
73 deploy a fixed rational approximation $r(v) = P(v)/Q(v)$ to $v^{-1/2}$ over $v \in [0.1, 10]$:

$$v = \frac{m(x)}{s_0}, \quad \text{RNorm}(x) = \frac{x}{\sqrt{s_0}} r(v). \quad (3)$$

74 Here rational means a ratio of polynomials. Projective coordinates carry each activation as a
75 numerator-denominator pair (N, D) representing N/D . Updating this pair preserves input-specific
76 rescaling without leaving the rational tensor representation. Training and deployment use r itself. The
77 exactness claims below therefore concern faithful evaluation of the deployed rational computation,
78 not uniform agreement with RMSNorm. Appendix A.3 quantifies that distinction.

79 2.2 Architecture and training

80 Both encoder variants follow the high-level VLA sequence of visual encoding, language and state
81 embedding, joint processing, and action decoding [1]. They use a 64×64 agent-view image, an
82 eight-dimensional robot state, a 26-token task vocabulary, and eight action-query tokens. Both share
83 a width-384, eight-block, twelve-head joint backbone and a linear head that maps each action-query
84 state to seven continuous action dimensions. The 20.1M ViT uses four bilinear-attention vision
85 blocks and has 20,137,352 learned parameters. The 19.2M convolutional variant instead uses three
86 stride-2 bilinear convolution stages that produce an 8×8 token grid. The historical capability records
87 use the convolutional variant. The runtime operator census and reduced-Gram screen use the ViT. The
88 released convolutional files additionally receive the joint-attention and rational-normalizer certificates
89 reported in Appendix A.3.

90 Training uses 66,984 demonstration frames, 40,000 AdamW updates, batch size 256, cosine decay,
91 bfloat16 arithmetic, and an exponential moving average. Closed-loop evaluation uses a 180° camera
92 rotation to match the demonstration convention, canonical LIBERO initial states, and fail-loud checks
93 that verify every requested state was loaded.

94 2.3 Mechanical certificate on the 20.1M ViT checkpoint

95 Architecture definitions can differ from deployed code. We therefore audit the real 20.1M ViT
96 189/200 checkpoint at runtime and reconstruct representative operations from saved weights.

Mechanical decomposability audit of the trained 189/200 checkpoint

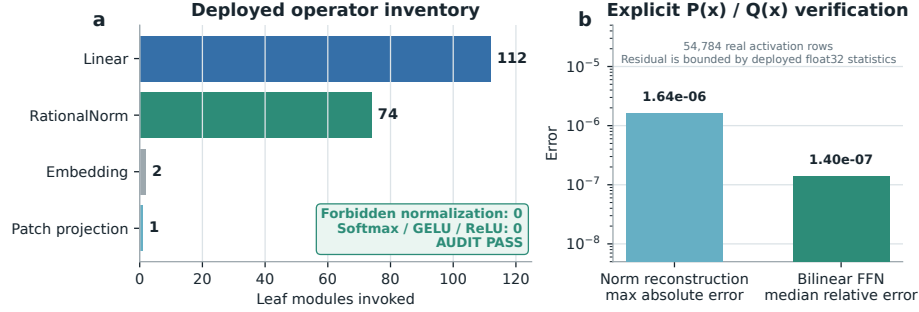


Figure 1: **Mechanical audit of the 20.1M ViT rational checkpoint.** The runtime path invokes 112 linear maps, 74 rational normalizers, two embeddings, and one linear patch projection. No softmax, standard activation, or ordinary normalization is present. Reconstruction errors are measured over 54,784 real activation rows.

The normalization reconstruction has maximum absolute error 1.64×10^{-6} . A bilinear FFN reconstruction has median relative error 1.40×10^{-7} . These residuals are bounded by the deployed module’s float32 statistic. The audit certifies operator membership and local reconstruction. It does not materialize one compact polynomial for the full eight-layer transformer.

3 The artifact is a capable closed-loop policy

3.1 Evaluation protocol

The historical aggregate records specify the following evaluation protocol for all ten LIBERO-Object tasks:

- 50 canonical trials per task
- a 280-step cap
- three independently trained convolutional seeds
- 500 rollouts per seed and 1,500 rollouts in total

Published OpenVLA and Diffusion Policy results are protocol-aligned references. They were not rerun in our codebase, so differences may still include implementation and training choices.

3.2 Results

Method	Parameters	Training seeds	Success
χ -VLA, rational conv	19.2M	3	93.7% $\pm$ 0.8%
χ -VLA, conv ensemble	$3 \times 19.2M$	$3 \rightarrow 1$	96.2%
OpenVLA-7B [1]	7B	3	88.4% $\pm$ 0.8%
Diffusion Policy [1]	not matched	3	92.5% $\pm$ 0.7%

Table 2: **Closed-loop success under aligned evaluation settings.** The single-model χ -VLA row is the mean $\pm$ sample standard deviation from surviving historical aggregate records. Their original per-episode files are unavailable. The ensemble row comes from a separate paired evaluation. The two external rows are historical references rather than current state-of-the-art claims.

The three historical convolutional aggregate records report 94.0%, 94.4%, and 92.8%. Their narrow spread is descriptive bounded evidence across independently trained checkpoints.

Separately, the paired ensemble evaluation re-evaluates the three members at 93.6%, 95.0%, and 93.2%, for a 93.9% mean. Elementwise prediction averaging reaches 96.2%. This is 2.3 points

116 above that paired mean and 1.2 points above its strongest member. The gain is largest on tomato
117 sauce, butter, and ketchup. The ensemble requires three complete forward passes and is not one
118 19.2M-parameter policy.

119 3.3 What this result does and does not establish

120 The result shows that the rational architecture supports high closed-loop specialist capability. It is an
121 existence claim for a compact tensor-decomposable policy, not a claim of broad robotic generalization
122 or state of the art.

123 4 Exact weight-derived structure through attention

124 4.1 Exact layerwise construction

125 Equation 2 contains two query-key products and one value stream. Expanding the linear maps gives
126 a coefficient table whose entries measure how combinations of query, key, and value coordinates
127 contribute to an output. No examples or fitted probe are needed to construct it. On a tiny four-
128 token, width-four, one-head layer, direct evaluation and an independent explicit polynomial agree to
129 2.00×10^{-15} . The internal core identity agrees to 1.24×10^{-14} .

130 The deployed attention also normalizes each query and key with Equation 3. Each normalization
131 multiplies all channels of one token by the same scalar rational factor. Those factors can be pulled
132 outside the bilinear dot products. The polynomial attention core is unchanged, while explicit
133 numerator and denominator factors carry the normalization. The rational extension agrees to $3.61 \times$
134 10^{-16} . We then reconstruct all four vision and eight joint attention modules in each of three
135 trained checkpoints, covering 384 heads in total. The worst relative ℓ_2 error is 2.56×10^{-7} and the
136 worst absolute difference is 2.23×10^{-5} . This residual comes from the deployed implementation’s
137 intentional float32 statistic, while the reconstruction itself uses the learned coefficients.

138 We separately certify the three released convolutional checkpoint files named in the historical
139 capability aggregate records. Their vision stacks contain no attention, so we independently rebuild all
140 eight joint modules and all twelve heads per module from the saved weights. Sixteen deterministically
141 selected cache inputs per checkpoint yield 384 module cases and 4,608 head-input cases. Every
142 output is finite and the worst relative ℓ_2 error is 6.36×10^{-16} , far below the frozen 10^{-6} gate. This
143 same-checkpoint certificate covers the joint-attention equation at each normalized layer input. It does
144 not rebuild the convolutional stem, feed-forward blocks, residual composition, final normalization, or
145 action head as one end-to-end contraction.

146 Naively materializing the real width-384 coefficient tensor would require approximately 3.2×10^{15}
147 values. We instead form an $O(d^2)$ reduced Gram. Like a covariance matrix, its leading eigenvectors
148 rank important directions without constructing the full table. It matches brute-force materialization to
149 2.13×10^{-14} at toy scale.

150 4.2 Trained policy result

151 The reconstruction certificate above spans three ViT checkpoints. The following subspace-ranking
152 experiment is a descriptive analysis of the seed-0 ViT checkpoint. We apply the exact weight-derived
153 calculation to all twelve heads in early, middle-late, and final backbone blocks 0, 6, and 7. Random
154 projector banks are sampled once, then fixed and reused for all examples. The weights choose the
155 subspace, while cached activations are used only afterward to score how well it preserves the layer’s
156 computation.

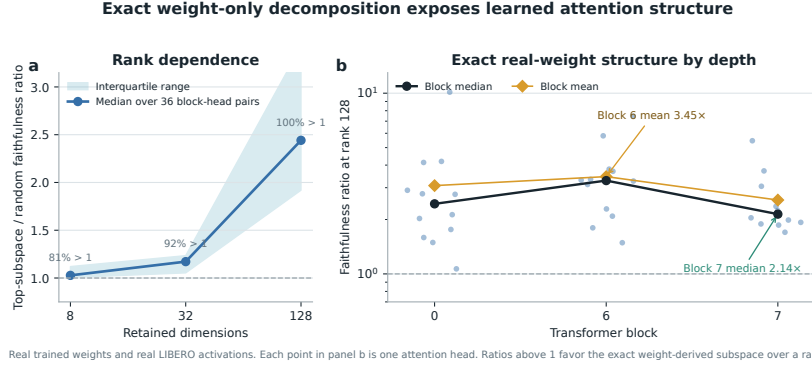


Figure 2: **Exact weight-derived attention structure.** Ratios above one favor the exact weight-derived subspace over a random equal-rank subspace. Each point in the right panel is one trained attention head.

At ranks 8, 32, and 128, respectively, 80.6%, 91.7%, and 100% of the 36 block-head pairs favor the exact subspace. Their median ratios are 1.028, 1.171, and 2.441. At rank 128, the mean ratio is 3.033 and the strongest head reaches 10.135. The rank-128 block means are 3.08, 3.45, and 2.56 for blocks 0, 6, and 7. The 36 block-head pairs are architectural units within one checkpoint, not independent model-level replicates.

This extends exact weight-only construction through individual attention layers. It is not an exact end-to-end decomposition of the complete policy. Composition across all eight blocks still faces rapid degree and representation growth.

5 Related work and limitations

Related work. OpenVLA and OpenVLA-OFT establish pretrained VLA and continuous-control baselines [1, 2]. Bilinear MLPs, χ -nets, and bilinear IMPALA policies show that multiplicative weights can expose low-rank structure and support subsequent controls [3–5]. Recent work also steers pretrained VLAs through activation directions [6]. Our distinction is the exact weight-derived construction in a policy designed as a rational tensor program. Tensor and polynomial networks provide the broader multilinear foundation [12–14]. VLA robustness benchmarks show that standard LIBERO success can coexist with weak language use [9–11].

Limitations and conclusion. The benchmark is one specialist suite, so it does not establish broad robotic generalization. Exact reconstruction is certified layer by layer and does not yet produce one compact symbolic contraction of the full policy. The released convolutional files support the joint-attention equation certificate and are named in the historical capability aggregate records. The reduced-Gram subspace screen uses separate ViT checkpoints. Evaluation uses fixed task instructions and does not establish robust instruction grounding. Within that scope, rational restrictions support strong LIBERO-Object control and exact trained-weight certificates. We do not claim a direct coefficient-edit interface.

Reproducibility. The anonymous artifact releases raw structural-certificate JSONs, immutable SHA-256 identities, and a standard-library one-command verifier. Historical Modal cache-backed analyses read dataset-native task metadata. The replacement Athena pipeline pins that metadata and maps dataset identifiers to official identifiers before defining task splits. A source-level audit compared all 271,996 cached frames across Object, Spatial, Goal, and LIBERO-10 with their pinned LeRobot revisions. Task identifiers, language joins, states, actions, and resized images matched exactly for every frame. Canonical-state checks fail loudly when requested states are unavailable.

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213 A Supporting results and decisions

214 A.1 Per-task ensemble behavior

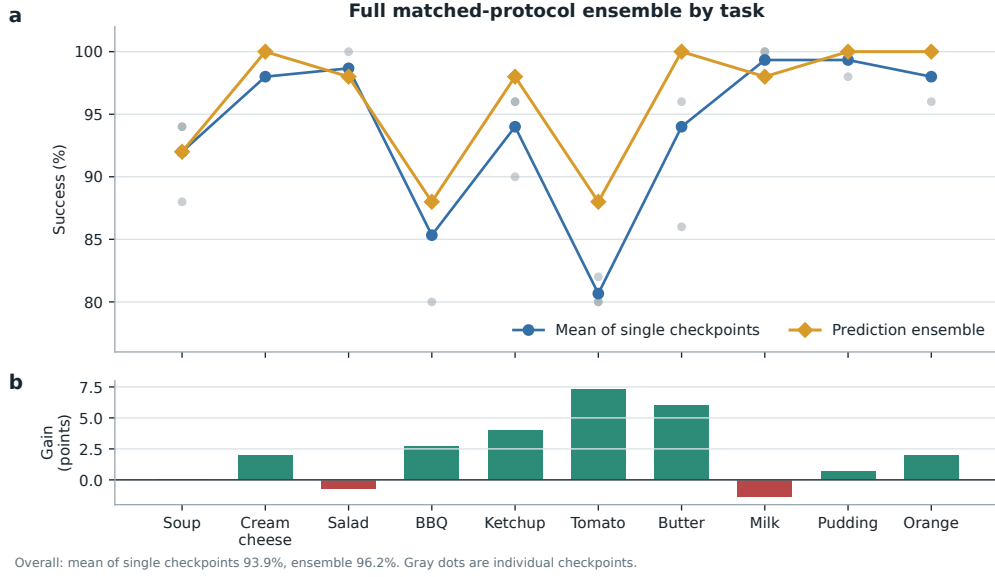


Figure 3: **Per-task effect of prediction averaging in the paired ensemble evaluation.** Gray points are the three convolutional checkpoints, evaluated with 50 canonical trials per task. Curves compare their taskwise mean with elementwise action-prediction averaging. The ensemble raises overall success from a 93.9% member mean to 96.2%, with its largest gains on tomato sauce, butter, and ketchup. It requires three forward passes and is not a single 19.2M-parameter policy.

215 A.2 Expanded attention screen

216 Within the same seed-0 ViT checkpoint, the corrected fixed-control attention analysis covers all
 217 eight blocks and twelve heads. At rank 128, 83/96 block-head pairs favor the exact weight-derived
 218 subspace. The median ratio is 1.94, the mean is 2.58, and blocks 2 and 3 provide useful depth
 219 controls. At ranks 8 and 32, only 54/96 and 60/96 pairs favor the exact subspace. The effect is
 220 therefore depth-dependent and strongest at the wider tested rank.

221 A.3 Numerical scope of rational conversion

222 The deployed degree- $[2/2]$ least-squares rational fit is exact as a rational computation even though it
 223 approximates RMS normalization. Its numerator has coefficients (5.32997, 7.53552, 0.327274) and
 224 the denominator has coefficients (1, 9.64827, 2.60664). Every denominator coefficient is positive,
 225 which certifies $Q(v) > 0$ for all $v \geq 0$. We instrument all 53 RationalNorm module instances,
 226 each invoked once per forward pass, on 4,096 deterministic inputs for each of three convolutional
 227 checkpoints. Across 544,542,720 activation rows, the minimum observed denominator is 1.0016 and
 228 no row is nonfinite. Comparing deployed float32 normalization with an otherwise identical pass that
 229 recomputes only RationalNorm scales in float64, the largest checkpoint-level primary metric, defined
 230 as the maximum of normalized- and physical-action NRMSE, is 2.414×10^{-6} against a frozen 10^{-3}
 231 gate.

232 The certificate also bounds what the operator approximates. Observed normalized mean-square ratios
 233 v span 1.66×10^{-4} to 17.17, outside the nominal fit interval $[0.1, 10]$. Across site-checkpoint pairs,
 234 the minimum coverages are 60.018% in that interval and 60.022% with rational-versus-exact-RMS
 235 scale error at most 3.4%. The frozen 99% secondary coverage gates therefore do not pass. The
 236 positive claim is therefore global absence of nonnegative denominator poles and cache-conditioned
 237 floating-point fidelity to the deployed rational operator, not uniform agreement with exact RMS
 238 scaling. Matched alternative-normalizer runtime overhead remains unmeasured.

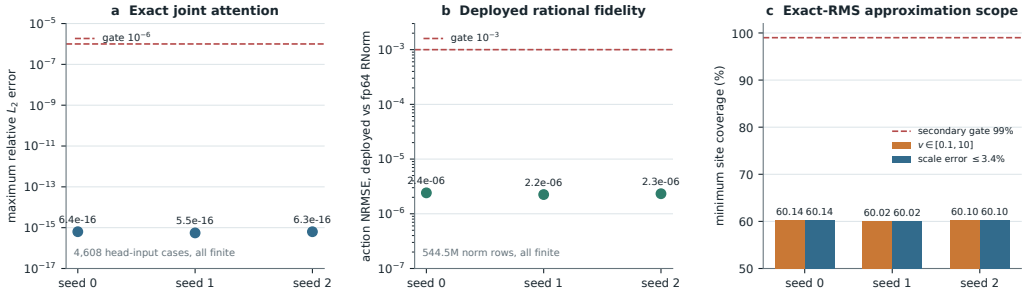


Figure 4: **Frozen structural certificates on the released convolutional files named in the historical capability aggregate records.** Panels (a) and (b) pass their primary gates with large margins. Panel (a) reconstructs all 4,608 audited joint-attention head-input cases. Panel (b) compares deployed float32 normalization with an otherwise identical pass that recomputes only RationalNorm scales in float64. Panel (c) reports the separately frozen exact-RMS approximation coverage, which does not pass its 99% secondary gate.